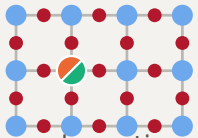
Pr and Rb both lower the O-vacancy cost in TiO_2; at the O-rich limit only Rb makes it spontaneous

Anatase, spin-polarised PBE+U, U(Ti-3d) = 3.5 eV · oxygen-rich reference $\mu_O = E(O_2)/2$ · neutral V_O, one dopant per 48-atom cell

❶ Build cell



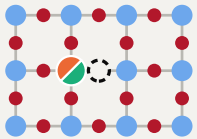
schematic

Ti O Pr/Rb

2×2×1 anatase, 48 atoms

One Ti → Pr^{3+} or Rb^{+} (separate cells)

❷ Remove O



V_O: removed O

First-shell O of the dopant

Neutral V_O releases

●● 2 e⁻

❸ Protocol

Γ-only relaxations, 40 Ry

$E_f = E(V_O) - E(\text{perfect}) + E(O_2)/2$

Δ(Pr–Rb) = 5.36 eV

✓ 2×2×2 k-mesh, 55, 70 Ry, D3: Δ changes ≤ 0.04 eV

! narrower smearing: Pr cells spin-polarise

Pr +0.99, Rb −4.13 eV, Δ = 5.11 eV

! U(O-2p): spin states of Pr, Rb parent cells change

E_f not recomputed

❹ Charge bookkeeping

○ hole h⁺ ● electron e⁻

formal (Kröger–Vink) counts

Pr^{3+}

dopant V_O net

○ + ●● → ● **1 e⁻ left**

extra e⁻ in Ti-3d conduction band; narrower smearing: partly on 4 Ti

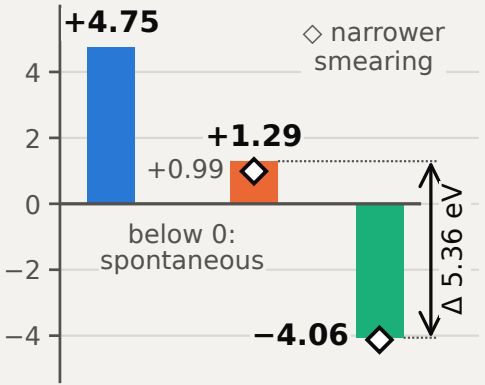
Rb^{+}

○○○ + ●● → ○ **1 h⁺ left**

residual h⁺ on O-2p: Löwdin 0.59 μ_B on one O bonded to Rb

Rb lies 1.27 Å off its ideal site, towards V_O

❺ Result: $E_f(V_O^0)$ (eV)



h⁺: formal holes before V_O removal

Å: cation radius; both rise together